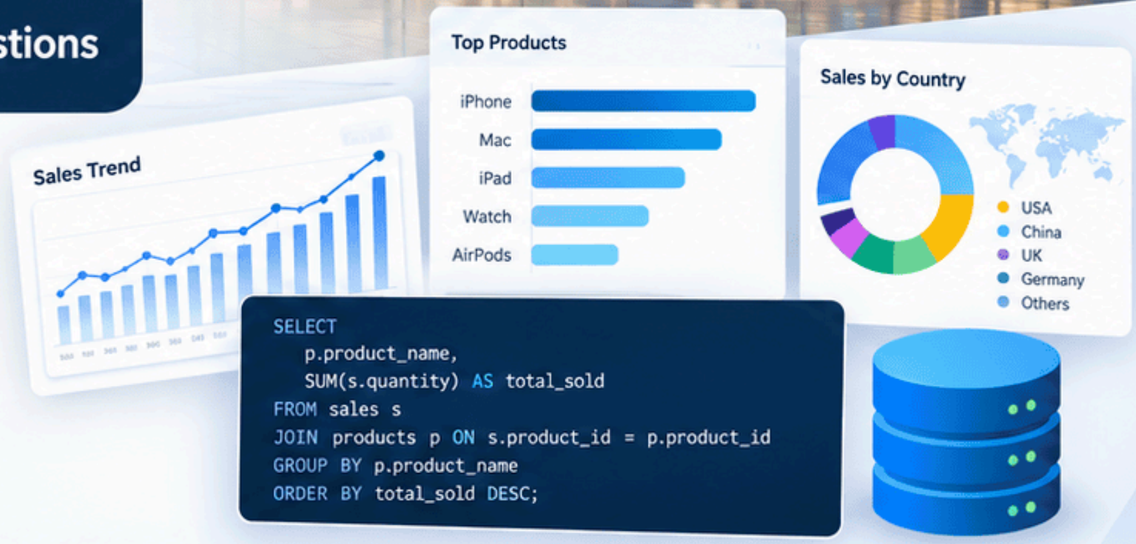
Apple Store Sales SQL Data Analysis

Real Data | Powerful Insights | Better Decisions

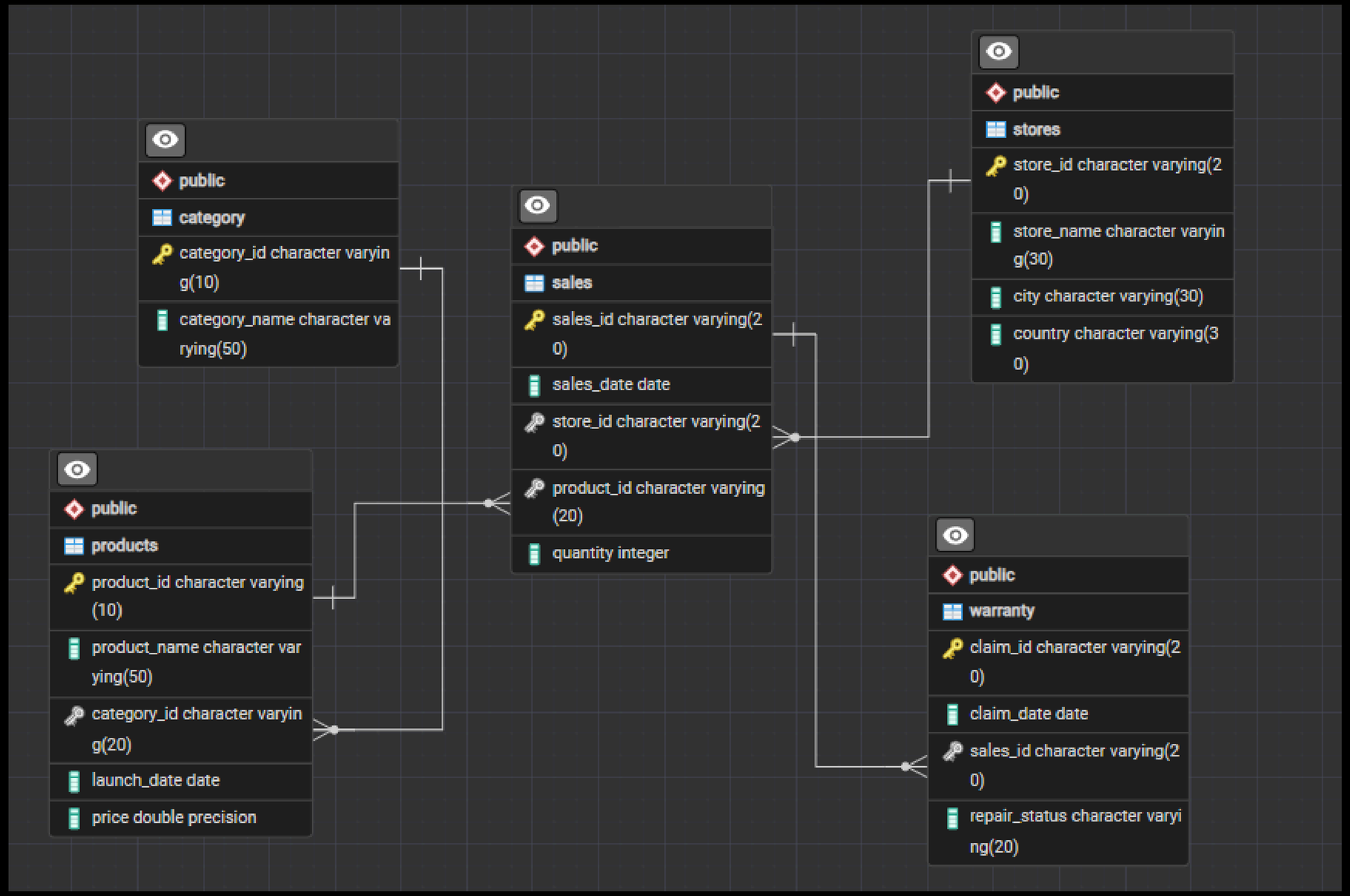
PostgreSQL | 1M+ Records | 21 Business Questions



PostgreSQL | SQL | GitHub | Data Analysis

5 Tables (1M+ Rows) | **Window Functions** (RANK, LAG) | **CTEs** (Common Table Expressions) | **Advanced Joins** (INNER, LEFT, RIGHT) | **Business Insights** (Real-World Analysis)

EDR


public
category
category_id character varying(10)
category_name character varying(50)
public
sales
sales_id character varying(20)
sales_date date
store_id character varying(20)
product_id character varying(20)
quantity integer
public
stores
store_id character varying(20)
store_name character varying(30)
city character varying(30)
country character varying(30)
public
products
product_id character varying(10)
product_name character varying(50)
category_id character varying(20)
launch_date date
price double precision
public
warranty
claim_id character varying(20)
claim_date date
sales_id character varying(20)
repair_status character varying(20)

Query | Query History

```
40
41
42  -- 1. Find the number of stores in each country.
43  SELECT country, COUNT(store_id) AS store_num
44  FROM stores
45  GROUP BY country
46  ORDER BY store_num DESC;
47
```

Data Output | Messages | Notifications

Showing rows: 1 to 19 | Page No: 1

	country character varying (30)	store_num bigint
1	United States	15
2	Australia	7
3	China	7
4	Japan	6
5	Canada	5
6	UAE	5
7	United Kingdom	4
8	France	4
9	Germany	3
10	Mexico	3
11	Singapore	3
12	Thailand	3
13	South Korea	2

Total rows: 19 | Query complete 00:00:00.136

```sql
-- 2. Calculate the total number of units sold by each store.
SELECT
    sl.store_id,
    st.store_name,
    SUM(sl.quantity) AS total_units
FROM sales AS sl
JOIN stores AS st
ON st.store_id = sl.store_id
GROUP BY 1,2
ORDER BY 3 DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 75 | Page No: 1 of 1

	store_id character varying (20)	store_name character varying (30)	total_units bigint
1	ST-56	Apple Southland	77795
2	ST-34	Apple Fukuoka	77787
3	ST-1	Apple Fifth Avenue	77689
4	ST-30	Apple Dubai Mall	77571
5	ST-24	Apple Kurfuerstendamm	77532
6	ST-39	Apple Taipei 101	77518
7	ST-75	Apple Beijing SKP	77482
8	ST-25	Apple Schildergasse	77385
9	ST-5	Apple SoHo	77186
10	ST-52	Apple Chadstone	77170
11	ST-12	Apple North Michigan Aven...	77152

Total rows: 75 | Query complete 00:00:00.847 | CRLF | Ln

Query | Query History

```
59  -- 3. Identify how many sales occurred in December 2023.
60  SELECT COUNT(*)
61  FROM sales
62  WHERE TO_CHAR(sales_date, 'MM-YYYY') = '12-2023';
63
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No

	count bigint
1	18076

Query | Query History

```sql
-- 4. Determine how many stores have never had a warranty claim filed.
SELECT COUNT(*) FROM stores AS st
WHERE st.store_id NOT IN (
                        SELECT DISTINCT store_id FROM sales AS sl
                        RIGHT JOIN warranty AS w
                        ON sl.sales_id = w.sales_id
                        );

```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No:

	count bigint
1	0

Query | Query History

```sql
-- 5. Calculate the percentage of warranty claims marked as "Warranty Rejected".
SELECT
    ROUND(COUNT(*)/
    (SELECT COUNT(*) FROM warranty) :: numeric
    *100,2) AS warranty_rejected_percentage
FROM warranty
WHERE repair_status = 'Rejected';
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page N

	warranty_rejected_percentage numeric
1	24.52

Query | Query History

```sql
-- 6. Identify which store had the highest total units sold in the last 2 year.
WITH abc AS
            (SELECT store_id, SUM(quantity) AS total_unit
            FROM sales
            WHERE sales_date >= (SELECT CURRENT_DATE - INTERVAL '2 year')
            GROUP BY store_id)
SELECT
    s.store_id, s.store_name, abc.total_unit
FROM stores AS s
LEFT JOIN abc
ON s.store_id = abc.store_id
ORDER BY abc.total_unit DESC
LIMIT 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	store_id [PK] character varying (20)	store_name character varying (30)	total_unit bigint
1	ST-2	Apple Union Square	3337

Query | Query History

```sql
-- 7. Count the number of unique products sold in the last 2 year.
-- Each Product sold.

SELECT
    product_id,
    COUNT(*) AS sold
FROM sales
WHERE sales_date >= (SELECT CURRENT_DATE - INTERVAL '2 year')
GROUP BY 1
ORDER BY 2 DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 89 | Page No: 1

	product_id character varying (20)	sold bigint
1	P-29	529
2	P-23	508
3	P-56	506
4	P-12	500
5	P-57	497
6	P-53	496
7	P-88	494
8	P-84	490
9	P-52	489
10	P-67	487

Total rows: 89 | Query complete 00:00:00.395

Query | Query History

```
116
117  -- 8. Find the average price of products in each category.
118  SELECT
119          p.category_id,
120          c.category_name,
121          AVG(p.price) AS avg_price
122  FROM products AS p
123  JOIN category AS c
124  ON c.category_id = p.category_id
125  GROUP BY 1,2
126  ORDER BY avg_price DESC;
127
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No:

	category_id character varying (20)	category_name character varying (50)	avg_price double precision
1	CAT-3	Tablet	1479.5
2	CAT-1	Laptop	1194.1
3	CAT-5	Wearable	1146.888888888889
4	CAT-2	Audio	1124.8181818181818
5	CAT-4	Smartphone	1037.5384615384614
6	CAT-10	Accessories	1030.142857142857
7	CAT-6	Streaming Device	996.6666666666666
8	CAT-7	Desktop	838.7
9	CAT-8	Subscription Service	823.2857142857143
10	CAT-9	Smart Speaker	734

Total rows: 10 | Query complete 00:00:00.176

Query | Query History

```
127
128
129  -- 9. How many warranty claims were filed Completed?
130
131  SELECT
132      COUNT(*)
133      -- EXTRACT(YEAR FROM claim_date) AS claim_year      -- Learn how to extract year from any date
134  FROM warranty
135  WHERE repair_status = 'Completed';
136
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1

	count bigint
1	7466

Query | Query History

```sql
-- 10. For each store, identify the best-selling day based on highest quantity sold.

SELECT * FROM(
        SELECT
            sl.store_id,
            TO_CHAR(sl.sales_date, 'Day') AS sale_day,
            SUM((p.price * sl.quantity)) AS net_price,
            RANK() OVER (PARTITION BY sl.store_id ORDER BY SUM((p.price * sl.quantity)) DESC) AS rank
        FROM sales AS sl
        LEFT JOIN products AS p
        ON sl.product_id = p.product_id
        GROUP BY 1,2) AS tb1
WHERE rank = 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 75 | Page No: 1 of 1

	store_id character varying (20)	sale_day text	net_price double precision	rank bigint
1	ST-1	Tuesday	12168968	1
2	ST-10	Tuesday	12109477	1
3	ST-11	Thursday	12769991	1
4	ST-12	Thursday	12088152	1
5	ST-13	Sunday	12418839	1
6	ST-14	Monday	12109068	1
7	ST-15	Thursday	11794485	1
8	ST-16	Tuesday	12067295	1

Total rows: 75 | Query complete 00:00:03.281 | CRLF | Ln

Query | Query History

```sql
-- 11. Identify the least selling product in each country for each year based on total units sold.

WITH tb1 AS (
    SELECT
        st.country,
        p.product_name,
        SUM(sl.quantity) AS total_unit,
        RANK() OVER(PARTITION BY st.country ORDER BY SUM(sl.quantity)) AS rank
    FROM sales AS sl
    JOIN stores AS st
    ON sl.store_id = st.store_id
    JOIN products AS p
    ON p.product_id = sl.product_id
    GROUP BY 1,2
    )
SELECT * FROM tb1
WHERE rank = 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 19 | Page No: 1

	country character varying (30)	product_name character varying (50)	total_unit bigint	rank bigint
1	Australia	MacBook Air (M1)	5516	1
2	Austria	Apple Music	663	1
3	Canada	Apple Pencil (1st Generati...	3798	1
4	China	iPhone 12	5343	1
5	Colombia	Beats Studio Buds	1428	1

Total rows: 19 | Query complete 00:00:01.182

Query | Query History

```sql
-- 12. Calculate how many warranty claims were filed within 180 days of a product sale.

WITH abc AS (
    SELECT
        sl.sales_id,
        sl.sales_date,
        sl.product_id,
        w.claim_date,
        (w.claim_date - sl.sales_date) AS diff_date
    FROM sales AS sl
    JOIN warranty AS w
    ON sl.sales_id = w.sales_id
    WHERE (w.claim_date - sl.sales_date) <= 180)
SELECT COUNT(*) FROM abc
WHERE diff_date >= 0;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	count bigint
1	3046

Query | Query History

```sql
-- 13. Determine how many warranty claims were filed for products launched in the last two years.

SELECT
    COUNT(*)
    -- sl.sales_id,
    -- sl.product_id,
    -- p.launch_date,
    -- w.claim_date,
    -- (claim_date - launch_date) AS diff_date
FROM sales AS sl
JOIN products AS p
ON sl.product_id = p.product_id
JOIN warranty AS w
ON w.sales_id = sl.sales_id
WHERE ((w.claim_date - p.launch_date) <=  730)
    AND ((w.claim_date - p.launch_date) >=  0);
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1

	count bigint
1	11369

Query | Query History

```sql
-- 14. List the months in the last three years where sales exceeded 5,000 units in the USA.
SELECT
    st.country,
    TO_CHAR(sl.sales_date, 'MM-YYYY'),
    SUM(sl.quantity) AS total_unit
FROM sales AS sl
JOIN stores AS st
ON st.store_id = sl.store_id
WHERE
    (country = 'United States')
    AND
    (sl.sales_date >= CURRENT_DATE - INTERVAL '3 YEAR')
GROUP BY 1,2
HAVING SUM(sl.quantity) > 5000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 15 | Page No: 1

	country character varying (30)	to_char text	total_unit bigint
1	United States	04-2024	19513
2	United States	10-2023	20238
3	United States	11-2024	7956
4	United States	09-2024	19203
5	United States	03-2024	20051
6	United States	11-2023	18926
7	United States	10-2024	19775

Total rows: 15 | Query complete 00:00:00.305

Query | Query History

```sql
-- 15. Identify the product category with the most warranty claims filed in the last two years.

SELECT
    c.category_name,
    COUNT(w.claim_id) AS total_claim
FROM warranty AS w
LEFT JOIN sales AS sl
ON w.sales_id = sl.sales_id
JOIN products AS p
ON p.product_id = sl.product_id
JOIN category AS c
ON c.category_id = p.category_id
WHERE w.claim_date >= CURRENT_DATE -INTERVAL '2 YEAR'
GROUP BY 1
ORDER BY total_claim DESC
LIMIT 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1

	category_name character varying (50)	total_claim bigint
1	Accessories	1026

Query | Query History

```sql
-- 16. Determine the percentage chance of receiving warranty claims after each purchase for each country.
SELECT
    *,
    ((total_claim :: numeric / total_unit :: numeric) * 100) AS risk
FROM
    (SELECT
        st.country,
        SUM(sl.quantity) AS total_unit,
        COUNT(w.claim_id) AS total_claim
    FROM sales AS sl
    JOIN stores AS st
    ON sl.store_id = st.store_id
    LEFT JOIN warranty AS w
    ON w.sales_id = sl.sales_id
    GROUP BY 1) AS tb1
ORDER BY 4 DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 19 | Page No: 1

	country character varying (30)	total_unit bigint	total_claim bigint	risk numeric
1	Austria	75965	448	0.58974527743039557700
2	Netherlands	76764	428	0.55755301964462508500
3	Taiwan	77518	419	0.54051962124925823700
4	UAE	381972	2046	0.53564135591090446400
5	Canada	381212	2035	0.53382369914902993600

Total rows: 19 | Query complete 00:00:00.913

Query History

```sql
-- 17. Analyze the year-by-year growth ratio for each store.
WITH tb1 AS
        (SELECT
            st.store_name,
            EXTRACT(YEAR FROM sl.sales_date) AS sales_year,
            SUM(sl.quantity * p.price ) AS current_sales
        FROM sales AS sl
        JOIN products AS p
        ON p.product_id = sl.product_id
        JOIN stores AS st
        ON st.store_id = sl.store_id
        GROUP BY 1,2),
    tb2 AS
        (SELECT
            tb1.*,
            LAG(current_sales, 1) OVER(PARTITION BY store_name ORDER BY sales_year) AS previous_sales
        FROM tb1)
SELECT
    tb2.*,
    ROUND(((tb2.current_sales - tb2.previous_sales) :: numeric / tb2.previous_sales :: numeric) * 100,2) AS growth
FROM tb2
WHERE (previous_sales IS NOT NULL)
    AND
    (sales_year <> 2024);        -- Current year(2024) is running that's we are ignoring it.
```

Data Output | Messages | Notifications

	store_name character varying (30)	sales_year numeric	current_sales double precision	previous_sales double precision	growth numeric
1	Apple Ala Moana	2021	16351838	16736679	-2.30
2	Apple Ala Moana	2022	16679728	16351838	2.01
3	Apple Ala Moana	2023	17286752	16679728	3.64
4	Apple Andino	2021	16657275	15873867	4.94
5	Apple Andino	2022	16807255	16657275	0.90
6	Apple Andino	2023	16452478	16807255	-2.11
7	Apple Antara	2021	16516324	16614977	-0.59
8	Apple Antara	2022	16625607	16516324	0.66
9	Apple Antara	2023	15789600	16625607	-5.03
10	Apple Beijing SKP	2021	16967537	17004025	-0.21
11	Apple Beijing SKP	2022	17293828	16967537	1.92
12	Apple Beijing SKP	2023	16793173	17293828	-2.89
13	Apple Beverly Center	2021	16086769	16876768	-4.68
14	Apple Beverly Center	2022	16653180	16086769	3.52
15	Apple Beverly Center	2023	16945007	16653180	1.75
16	Apple Bondi	2021	16883063	16887187	-0.02

Total rows: 207 | Query complete 00:00:03.317

Query | Query History

```sql
-- 18. Calculate the correlation between product price and warranty claims
-- for products sold in the last five years, segmented by price range.
SELECT
    CASE
        WHEN p.price < 500 THEN 'Less Expensive'
        WHEN p.price BETWEEN 500 AND 1000 THEN 'Mid Range Expensive'
        ELSE 'Very Expensive Product'
    END,
    COUNT(w.claim_id) AS claimed_item
FROM warranty AS w
LEFT JOIN sales AS sl
ON sl.sales_id = w.sales_id
JOIN products AS p
ON p.product_id = sl.product_id
WHERE sl.sales_date >= CURRENT_DATE - INTERVAL '5 YEAR'
GROUP BY 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1

	case text	claimed_item bigint
1	Less Expensive	3321
2	Mid Range Expensive	5355
3	Very Expensive Prod...	11021

Query History

```sql
-- 19. Identify the store with the highest percentage of "Completed" claims relative to total claims filed.
WITH tb1 AS
    (SELECT
        st.store_id, COUNT(w.claim_id) AS Completed_Claim
    FROM warranty AS w
    JOIN sales AS sl ON w.sales_id = sl.sales_id
    JOIN stores AS st ON st.store_id = sl.store_id
    WHERE w.repair_status = 'Completed'
    GROUP BY 1),
tb2 AS
    (SELECT
        st.store_id, COUNT(w.claim_id) AS All_Claim
    FROM warranty AS w
    JOIN sales AS sl ON w.sales_id = sl.sales_id
    JOIN stores AS st ON st.store_id = sl.store_id
    GROUP BY 1)
SELECT
    tb1.store_id, st.store_name, tb1.completed_claim, tb2.all_claim,
    ROUND((tb1.completed_claim :: numeric / tb2.all_claim :: numeric) * 100, 2) AS completed_claim_percentage
FROM tb1 JOIN tb2
ON tb1.store_id = tb2.store_id
JOIN stores AS st
ON st.store_id = tb1.store_id
ORDER BY 5 DESC;
```

Data Output | Messages | Notifications

Showing row

	store_id [PK] character varying (20)	store_name character varying (30)	completed_claim bigint	all_claim bigint	completed_claim_percentage numeric
1	ST-65	Apple Iconsiam	114	377	30.24
2	ST-59	Apple Mall of the Emirates	110	365	30.14
3	ST-16	Apple Regent Street	127	437	29.06
4	ST-55	Apple Highpoint	118	413	28.57
5	ST-40	Apple Causeway Bay	105	368	28.53
6	ST-8	Apple Pioneer Place	108	381	28.35
7	ST-62	Apple Via Santa Fe	116	414	28.02
8	ST-48	Apple Sainte-Catherine	108	386	27.98
9	ST-69	Apple The Dubai Mall	116	419	27.68
10	ST-33	Apple Marunouchi	109	395	27.59
11	ST-25	Apple Schildergasse	111	404	27.48
12	ST-57	Apple Yas Mall	110	405	27.16
13	ST-31	Apple Shinjuku	120	442	27.15
14	ST-71	Apple Yeouido	102	376	27.13
15	ST-35	Apple Shanghai IFC	103	381	27.03

Total rows: 75 | Query complete 00:00:00.570

Query | Query History

```sql
-- 20. Write a query to calculate the monthly running total of sales for each store
-- over the past four years and compare trends during this period.
WITH tb1 AS
    (SELECT
        sl.store_id,
        EXTRACT(YEAR FROM sl.sales_date) AS sales_year,
        EXTRACT(MONTH FROM sl.sales_date) AS sales_month,
        SUM(sl.quantity * p.price ) AS total_revenue
    FROM sales AS sl
    LEFT JOIN products AS p ON sl.product_id = p.product_id
    GROUP BY 1,2,3
    ORDER BY 1,2,3)
SELECT
    store_id, sales_year, sales_month, total_revenue,
    SUM(total_revenue) OVER(PARTITION BY store_id ORDER BY sales_year, sales_month) AS running_total
FROM tb1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1

	store_id character varying (20)	sales_year numeric	sales_month numeric	total_revenue double precision	running_total double precision
1	ST-1	2020	1	1519710	1519710
2	ST-1	2020	2	1513077	3032787
3	ST-1	2020	3	1456142	4488929
4	ST-1	2020	4	1432050	5920979
5	ST-1	2020	5	1408946	7329925
6	ST-1	2020	6	1398598	8728523

Total rows: 4425 | Query complete 00:00:04.039

Query | Query History

```sql
-- 21. Analyze product sales trends over time, segmented into key periods:
-- from launch to 6 months, 6-12 months, 12-18 months, and beyond 18 months.

WITH tab1 AS
    (SELECT
        p.product_name,
        CASE
            WHEN sl.sales_date BETWEEN p.launch_date AND p.launch_date + INTERVAL '6 MONTH' THEN '0-6'
            WHEN sl.sales_date BETWEEN p.launch_date + INTERVAL '6 MONTH' AND p.launch_date + INTERVAL '12 MONTH' THEN '6-12'
            WHEN sl.sales_date BETWEEN p.launch_date + INTERVAL '12 MONTH' AND p.launch_date + INTERVAL '18 MONTH' THEN '12-18'
            ELSE '18+'
        END AS plc,
        SUM(sl.quantity) AS total_qty_sale
    FROM sales AS sl
    JOIN products AS p
    ON p.product_id = sl.product_id
    GROUP BY 1,2)
SELECT * FROM tab1
ORDER BY 1,
        CASE
            WHEN plc = '0-6' THEN 1
            WHEN plc = '6-12' THEN 2
            WHEN plc = '12-18' THEN 3
            ELSE 4
        END;
```

Data Output | Messages | Notifications

	product_name character varying (50)	plc text	total_qty_sale bigint
1	AirPods (2nd Generation)	0-6	6650
2	AirPods (2nd Generation)	6-12	6635
3	AirPods (2nd Generation)	12-18	6615
4	AirPods (2nd Generation)	18+	44681
5	AirPods (3rd Generation)	0-6	6536
6	AirPods (3rd Generation)	6-12	6652
7	AirPods (3rd Generation)	12-18	6878
8	AirPods (3rd Generation)	18+	44036
9	AirPods Max	0-6	6606
10	AirPods Max	6-12	6881
11	AirPods Max	12-18	6511
12	AirPods Max	18+	44846
13	AirPods Pro	0-6	1394
14	AirPods Pro	18+	63349
15	AirPods Pro (2nd Generatio...	0-6	6788
16	AirPods Pro (2nd Generatio...	6-12	6383
17	AirPods Pro (2nd Generatio...	12-18	6612

Total rows: 326 | Query complete 00:00:03.019